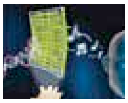
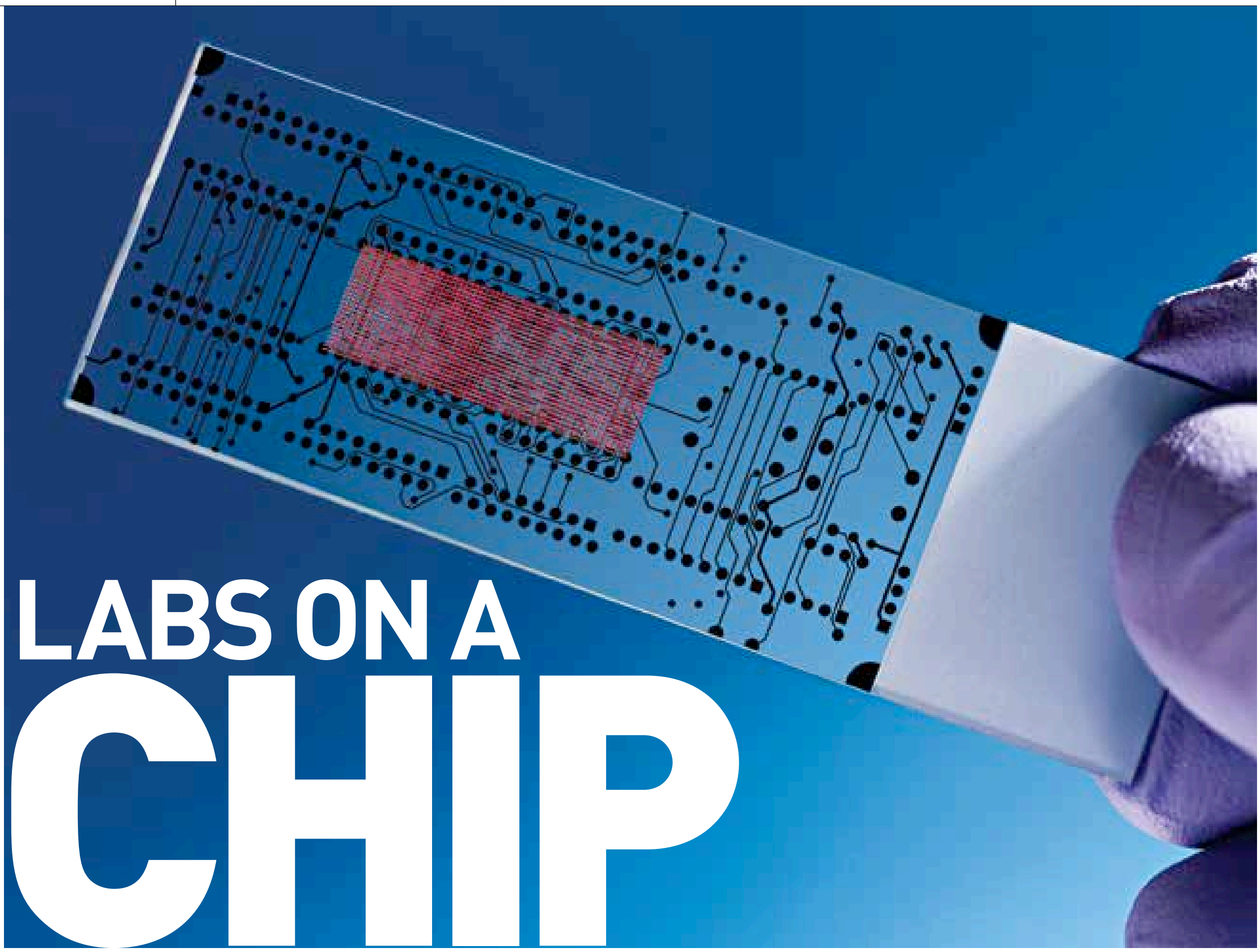


Let skin plays music
A team of researchers have presented an wearable tech of trasparent speakers that turns your skin into speakers. **https://dgit.in/oct18-72-1**

Superthin antennas
Researchers created 2D metallic flake antennas which can be printed onto electronics to bring them online. **https://dgit.in/oct18-72-2**



LABS ON A CHIP

An emerging category of hardware promises to revolutionise healthcare around the world, improve the performance of computers, and act as passive power systems for robots.

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Labs on a chip typically have small amounts of microfluids deposited on silicon chips to rapidly conduct experiments. The microfluidic devices are a kind of microelectromechanical (MEMS) devices. At times they may have living cells lining the silicon wafers. Because of the low amounts of fluids used in the devices, they have many advantages over traditional laboratories. The tests can be conducted quickly, it takes less energy to heat the fluids, only small amounts are needed for testing, and the cost of producing the chips are very low. For example, Professor Aydogan Ozcan from UCLA has created a flow cytometer that fits in the palm of a human hand and costs $5 to product. The device can be used to diagnose diseases, including AIDS, and track its progress. Traditional cytometers are bulky and cost tens of thousands of dollars. So it is quite obvious how using labs on a chip can be huge advantage for R&D purposes.

BRAIN ON A CHIP

You might have heard of a new type of computer chips – neuromorphic chips that mimic the internal structures of the human brain. These chips are mere approximations that are nowhere close to realising the actual complexity of the human brain. There are 100 billion neurons in there. Then there are synapses – 100 trillion of them to allow the neurons to talk to each other through neurotransmitters. Instead of working in binary, the neuromorphic chips work by assigning weights, or a gradient signals, in an analog manner. This would allow tiny neuromorphic chips to handle massive amounts of parallel processing, which have traditionally required massive supercomputers. The problem so far has been precisely replicating the microarcitecture of the human brain on small chips. Engineers from MIT have created